Patrol Log — FDL Brainstorm Plan

> **DRAFT — review artifact only.** This file exists so you can review the design in ultrapla na/ browser before blueprints are generated. The terminal artifact of the brainstorm is the set of blueprint YAML files in `blueprints/`. Once those are created and validated, this file can be deleted or archived. Do not edit the blueprints by editing this file.

Source of truth for review: this document. **Next step after approval**: `/fdl-create` per blueprint → writes YAML → `node scripts/validate.js → node scripts/completeness-check.js`.

Cross-cutting concerns (apply to all blueprints)

- **Auth prerequisite.** Every blueprint except `auth/patroller-login` requires an authenticated session with a valid device token.
- **Append-only + cryptographic seal.** Patrol records (`workflow/commence-patrol` and `workflow/stand-down-patrol`) are append-only while `state = active` and cryptographically sealed on transition to `stood_down`. Any post-seal edit must be recorded as a separate correction record with actor and reason — never as a silent mutation.
- **POPIA audit log.** Every read of `residents-directory` (search term, result count, tap-to-call target), every search of `members-directory`, and every write to a patrol record is audit-logged with actor call sign, device id, timestamp, and ip.
- **SARS logbook alignment.** Patrol records carry `start_location`, `end_location`, `odometer_start`, `odometer_end`, `distance_km` (computed = end − start), `sars_purpose`, `sars_compliant` (boolean). Records where `sars_compliant = false` are still retained but are excluded from tax-claim reports.
- **Location privacy.** Live location heartbeats live only in an ephemeral cache while the patrol is `active`. They are purged from the live cache within 5 s of stand-down. The full polyline persists on the sealed patrol record for audit only (not live).
- **Anti-scraping.** All directory reads (`residents`, `members`, `emergency`) are rate-limited to 30 queries/min/user.
- **Device-bound sessions.** Login is WhatsApp-style: first login issues a device token stored on the device; subsequent app opens auto-resume. Manual logout or admin force-revoke ends the session.

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1 — `auth/patroller-login`

Problem. Only authorized CPF patrollers may log patrols or see resident/member PII. An unauthenticated or deactivated user must never reach the home screen.

Fields

Field	Type	Required	Notes
`call_sign`	string	yes	Uppercase alphanumeric, e.g. `WV46`
`password`	string	yes	Plain at submit; hashed argon2id at rest
`access_level`	enum	—	`call_centre_agent`, `patroller`, `sector_lead`, `admin`
`organization`	string	—	e.g. `CPF`
`sector`	string	—	e.g. `Wierdabrug Sector 1`
`province`	string	—	e.g. `Gauteng`
`device_id`	string	yes	Issued/verified at login
`device_token`	string	—	Long-lived, rotatable, returned on success

Success outcomes

- Given valid `call_sign` + `password` + a fresh `device_id` → session established, `device_token` issued and returned, profile fields loaded onto home dashboard (matches screen 2 — `Access Level: Call Centre Agent / Wierdabrug Sector 1 / CPF / Gauteng`).
- Given a previously issued `device_token` still valid on app open → auto-resume to home dashboard without re-prompting.

Failure outcomes

Code	Trigger
`INVALID_CREDENTIALS`	call sign or password wrong
`ACCOUNT_INACTIVE`	patroller record flagged inactive by admin
`MISSING_INPUT`	call sign or password empty
`RATE_LIMITED`	>5 attempts per 15 min per call sign or ip
`LOCKED_OUT`	10 consecutive failures → 30 min lockout
`DEVICE_REVOKED`	admin has revoked this `device_id`

Security rules

- MUST: passwords hashed with argon2id; never stored plaintext.
- MUST: rate limit 5 attempts / 15 min per `(call_sign, ip)` pair.

- MUST: lockout after 10 consecutive failures; auto-unlock after 30 min.
- MUST: `device_token` is opaque, revocable server-side, rotatable.
- SHOULD: audit log every attempt (success + failure) with call sign, ip, device id.
- MAY: MFA for `admin` access level (future — explicitly deferred).

Related: none (this is the entry point for all others).

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2 — `workflow/commence-patrol`

Problem. A CPF must know in real time *who* is on patrol, *where* they started, *what* mode (foot / vehicle / static), *what* vehicle, and *when* — for dispatch, incident response, SARS logbook, and duty-roster compliance. Without a commence record, nobody can prove later that a given patroller was on duty at a given time.

Fields

Field	Type	Required	Notes
`patrol_id`	uuid	yes	Server-generated
`primary_patroller_call_sign`	string	yes	From session
`joined_patroller_call_signs`	array	no	May be empty
`joined_patroller_start_times`	map<call_sign, timestamp>	—	Server-set per joined patroller
`patrol_type`	enum	yes	`foot`, `vehicle`, `static`
`patrol_vehicle`	string	conditional	Required when `patrol_type = vehicle`
`odometer_start`	number	conditional	Required when `patrol_vehicle` present; ≥ last recorded odometer for that vehicle
`start_time`	timestamp	yes	Server-set; never client-controlled
`start_location`	`object {lat, lng, accuracy_m, captured_at}`	no	Captured from device GPS; missing → `sars_compliant = false`

	`captured_at}`		`false`
`sars_purpose`	string	yes	Defaults to `"CPF sector patrol"`
`sars_compliant`	boolean	yes	`true` iff `start_location` present and `accuracy_m ≤ 100`
`sector`	string	yes	Derived from primary patroller's profile
`state`	enum	yes	`active` (only value at commence)

Success outcomes

- Given an authenticated `patroller`, `sector_lead`, or `admin` with no active patrol, AND valid `patrol_type`, AND — when `patrol_type = vehicle` — a valid `patrol_vehicle` not already on another active patrol, AND `odometer_start` ≥ the vehicle's last recorded reading → a new patrol record is created with `state = active`, `start_time = now` (server), `start_location` captured from device, any joined patroller call signs linked, `sars_compliant` set according to location quality.
- Given joined patrollers selected → each gets their own `start_time` in `joined_patroller_start_times` (enables independent stand-down per screen 4).
- Given the record is written → the home screen shows `Stand down from patrol` as the active option.

Failure outcomes

Code	Trigger
`UNAUTHORIZED`	Access level `call_centre_agent` cannot commence (dispatch role)
`ALREADY_ON_PATROL`	Primary is already on an active patrol
`INVALID_PATROL_TYPE`	`patrol_type` not in enum
`VEHICLE_REQUIRED`	`patrol_type = vehicle` but `patrol_vehicle` missing
`INVALID_VEHICLE`	Vehicle not registered or not available for this sector
`VEHICLE_IN_USE`	Vehicle already on another active patrol
`ODOMETER_START_INVALID`	Negative, non-numeric, or lower than last recorded reading
`JOINED_PATROLLER_UNAVAILABLE`	A selected joined patroller is on a different active patrol

`JOINED_PATROLLER_UNAVAILABLE`	or is inactive

Security rules

- MUST: only `patroller`, `sector_lead`, `admin` may commence. `call_centre_agent` is dispatch-only and is rejected.
- MUST: `start_time` is server-assigned, never client-supplied.
- MUST: vehicle allocation is atomic (SELECT FOR UPDATE or equivalent) to prevent double-booking under race.
- MUST: patrol record is append-only while `state = active`.
- SHOULD: audit log with `patrol_id`, `primary_call_sign`, `device_id`.

Related: `auth/patroller-login`.

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3 — `workflow/stand-down-patrol`

Problem. Every patrol that commenced must be closed out so duty logs balance, SARS-valid KMs are recorded against the right vehicle, and the vehicle becomes available for the next crew. Stand-down works per-individual: a joined patroller can leave early without ending the patrol, and — per the user's explicit requirement — the primary can stand down while joined patrollers continue under a newly nominated primary.

Fields

Field	Type	Required	Notes
`patrol_id`	uuid	yes	Must reference an `active` patrol
`actor_call_sign`	string	yes	From session
`role`	enum	yes	`primary` or `joined` — derived from patrol record
`odometer_end`	number	conditional	Required when `role = primary` and patrol has a vehicle; must be > `odometer_start`
`distance_km`	number	computed	`odometer_end - odometer_start`, server-computed, not user-entered
`end_time`	timestamp	yes	Server-set
`end_location`	`object {lat, lng, accuracy_m, captured_at}`	no	Like start_location: missing or inaccurate → flips `sars_compliant` to false

`reason`	enum	no	`shift_end`, `emergency`, `vehicle_issue`, `personal` (optional dropdown)
`handoff`	object	no	`{new_primary_call_sign, continue_vehicle: bool, new_vehicle?}` — only when primary nominates a successor
`record_seal_hash`	string	yes	Cryptographic hash over the full record written at stand-down

Success outcomes — three distinct transitions

3a — Joined patroller stands down early

- Given a joined patroller on an active patrol taps Stand Down → their personal `end_time` is recorded in the patrol record, they are removed from the active crew, the patrol's `state` stays `active`, the primary keeps driving. No odometer required for joined patrollers.

3b — Primary stands down, patrol closes

- Given the primary taps Stand Down, enters `odometer_end`, provides `end_location`, and does NOT nominate a handoff → the patrol transitions `active → stood_down`, `end_time = now`, `distance_km` is computed, vehicle is released, record is sealed (`record_seal_hash` written), any still-active joined patrollers are auto-stood-down with the same `end_time`.

3c — Primary stands down with handoff to new primary

- Given the primary taps Stand Down, enters `odometer_end`, provides `end_location`, AND nominates a joined patroller in `handoff.new_primary_call_sign`, AND that nominee is eligible (active account, correct access level, on this patrol, not already primary elsewhere) → the current patrol closes atomically (sealed with its KMs), AND a new patrol record is commenced with the nominee as primary, the same or a new vehicle, the nominee's own `odometer_start` (re-prompted), and remaining joined patrollers transferred. Continuous coverage, clean audit chain.

Failure outcomes

Code	Trigger
`NOT_ON_PATROL`	Actor is not part of any active patrol
`ALREADY_STOOD_DOWN`	Idempotent guard — double-tap no-op
`ODOMETER_END_REQUIRED`	Primary stand-down on a vehicle patrol without `odometer_end`

`ODOMETER_END_LESS_THAN_START`	`odometer_end < odometer_start` — physical impossibility
`ODOMETER_END_INVALID`	Non-numeric or absurdly large
`HANDOFF_NEW_PRIMARY_INVALID`	Nominee inactive, wrong access level, already primary elsewhere, not on this patrol
`HANDOFF_NO_CANDIDATES`	Handoff requested but no eligible joined patrollers remain
`UNAUTHORIZED`	Actor trying to stand down another patroller without admin/sector_lead role

Security rules

- MUST: actor must be a member of the patrol being stood down, OR have `sector_lead` / `admin` access level. Admins may force-close abandoned patrols with `reason` logged.
- MUST: `end_time` and `distance_km` are server-assigned. The UI field "KM's Traveled" on screen 4 becomes **read-only / computed** from `odometer_end - odometer_start`.
- MUST: idempotency — same `(patrol_id, actor_call_sign)` stand-down is a no-op if already processed.
- MUST: patrol record is sealed at stand-down; `record_seal_hash` is written over all fields. Any downstream edit must be a separate correction record.
- SHOULD: audit log every stand-down, especially handoffs (who handed to whom, why).

Related: `auth/patroller-login`, `workflow/commence-patrol`.

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4 — `data/hotspots-map`

Problem. Patrollers and dispatch need to see *where* incidents are concentrating over a chosen time window so patrol routes can be planned to intercept rather than react. A spreadsheet of coordinates is useless in the field; the map makes clustering immediately visible (screen 5 shows ~14 orange pins clustered in Valhalla and Glen Lauriston).

Fields (this is a query, not an entity)

Field	Type	Required	Notes
`period`	enum	yes	`today`, `7d`, `30d`, `90d` — preset range from "From period" / "To period" dropdowns
Response: `incidents`	array	—	`{lat, lng, incident_id, type, severity, occurred_at}` per match

Success outcomes

Success outcomes

- Given a valid `period` → all incidents from the external incidents API with `occurred_at` in that window are returned as geo-points; the map renders them with clustering where density is high.
- Given no incidents match → empty-state map renders with "No incidents in this period" message (not an error).
- Given a pin is tapped → incident summary shows type, time, severity. Full details require elevated access.

Failure outcomes

Code	Trigger
`INVALID_PERIOD`	Period not in the enum
`INCIDENTS_API_UNAVAILABLE`	Upstream incidents service down → fall back to cached data if available, degrade to list view if not
`MAP_PROVIDER_UNAVAILABLE`	Tile provider unavailable → show list-only view
`UNAUTHENTICATED`	No valid session

Security rules

- MUST: read-only; this blueprint never writes.
- MUST: any authenticated patroller sees any sector's hotspots (per user's explicit decision — no sector scoping on hotspots).
- MUST: cache results for 5 minutes per `period` to reduce load on the incidents API.
- SHOULD: redact incident details for types marked sensitive (domestic violence, minors) unless viewer has `sector_lead` + clearance.
- SHOULD: exact coordinates (per user decision) — no fuzzing.

Related: `auth/patroller-login`, `integration/incidents-api` (external, contract-only — not brainstormed in this round).

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5 — `data/residents-directory`

Problem. Mid-patrol, a CPF member needs to contact a resident at a specific address — to verify an alarm, report damage, or ask about a suspicious vehicle. A spreadsheet is too slow; the directory must be searchable and phone-callable in one tap, scoped to the patroller's sector.

Fields

Field	Type	Required	Notes
`search_term`	string	no	Min 2 chars when present
`sector`	string	—	Derived from session
Response item: `resident_id`	uuid	—	Not displayed; used for audit
Response item: `name`	string	—	Displayed
Response item: `phone`	string	—	Displayed; tap-to-call
Response item: `address`	string	—	Displayed verbatim (matches mockup)

Success outcomes

- Given an authenticated patroller enters a search term → residents in their sector matching on name, phone, or address are returned (three-line card matches screen 6).
- Given no search term → full resident list for the sector returned, paginated.
- Given a card is tapped → native dialer opens with the phone number; the tap event is audit-logged.

Failure outcomes

Code	Trigger
`UNAUTHORIZED`	Not authenticated
`NO_RESULTS`	Zero matches (rendered as empty state, not error toast)
`SEARCH_TERM_TOO_SHORT`	<2 chars when a term is supplied
`RATE_LIMITED`	>30 queries/min per user — anti-scraping
`SYNC_UNAVAILABLE`	Resident store offline → fall back to last known encrypted local cache if present

Security rules

- MUST: authenticated patroller only.
- MUST: sector-scoped — patrollers only see residents in their assigned sector (POPIA data minimization).
- MUST: rate limit 30 searches / min per user.
- MUST: audit log every search (term, user, timestamp) AND every tap-to-call (resident_id, user, timestamp) for POPIA access records.
- MUST: response never includes ID number, birthdate, or any PII beyond `name`, `phone`, `address`, even if present in the source store.

- SHOULD: on-device cache encrypted at rest, wiped on logout.
- SHOULD: server cache 5 min per `(sector, term)` tuple.

Related: `auth/patroller-login`.

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6 — `data/members-directory`

Problem. A patroller needs to reach a fellow CPF member by phone during an operation — to coordinate backup, hand off an incident, or reach a specific officer who has context on a repeat caller. Additionally, per user decision, next-of-kin contacts for each member live on this directory (not on the emergency list).

Fields

Field	Type	Required	Notes
`search_term`	string	no	Min 2 chars when present
Response item: `member_id`	uuid	—	Audit only
Response item: `name`	string	—	Displayed
Response item: `call_sign`	string	—	Displayed (e.g. `WC29`)
Response item: `phone`	string	—	Displayed; tap-to-call
Response item: `address`	string	—	Displayed verbatim (matches mockup)
Response item: `sector`	string	—	Displayed
Response item: `access_level`	enum	—	Displayed as role badge
Response item: `is_on_duty`	boolean	—	Computed by joining against live-patroller-map cache
Response item: `next_of_kin`	array	—	`{name, relationship, phone, alternate_phone}` — expandable section

Success outcomes

- Given an authenticated patroller enters a search term → members in their sector matching name, call sign, or phone are returned name, call sign, or phone are returned.

- Given no search term → full sector members list returned, paginated, sorted with on-duty members first.
- Given a card is tapped → dialer opens.
- Given the next-of-kin section is expanded → NoK entries are shown; tap-to-call on a NoK number is logged against the current active patrol if one exists.

Failure outcomes

Code	Trigger
`UNAUTHORIZED`	Not authenticated
`NO_RESULTS`	Zero matches
`SEARCH_TERM_TOO_SHORT`	<2 chars when supplied
`RATE_LIMITED`	>30/min
`MEMBER_INACTIVE_HIDDEN`	Suspended members silently excluded (soft filter, not error surfaced to user)

Security rules

- MUST: authenticated CPF members only; no cross-CPF visibility.
- MUST: sector-scoped for viewing (same rule as residents).
- MUST: inactive/suspended members excluded from all results.
- MUST: rate limit 30 searches/min.
- MUST: audit log all searches and all tap-to-call events (including NoK calls).
- MUST: directory is **read-only** for all users. Profile editing is a separate follow-up blueprint (`data/member-profile-admin-edit`) restricted to admins of the same sector as the target member.
- SHOULD: `is_on_duty` joined from the live-patroller-map ephemeral cache; stale-indicator if no heartbeat in >2 min.

Related: `auth/patroller-login`, `data/live-patroller-map`.

Follow-up blueprint noted: `data/member-profile-admin-edit`.

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7 — `data/emergency-contacts-directory`

Problem. When a patroller is confronting a real incident — a fire, a gunshot victim, a barricaded suspect — they need to reach the right emergency *service* in under three seconds. The app must present a tiny, curated, sector-aware list of services that works **even when the network is bad**, because emergencies often coincide with bad reception.

The list is for services (SAPS station, ambulance, fire brigade, armed response, hospital, ops room, vet, tow, poison control) — **not** civilian next-of-kin. Per user decision, next-of-kin lives on the Members list.

Fields

Field	Type	Required	Notes
`service_id`	uuid	—	
`name`	string	yes	e.g. `Valhalla SAPS`
`service_type`	enum	yes	`police`, `ambulance`, `fire`, `armed_response`, `hospital`, `ops_room`, `vet`, `tow`, `poison_control`, `other`
`primary_number`	string	yes	
`secondary_number`	string	no	After-hours line, specific operator, etc.
`address`	string	no	Tappable to open maps
`sector`	array	no	Sectors this service covers (an ambulance covers many)
`verified_at`	timestamp	yes	Drives the stale-verification warning
`priority`	integer	yes	Sort order, admin-tunable
`sensitive`	boolean	no	If true, hidden from non-patroller access levels

Success outcomes

- Given an authenticated patroller opens the screen → the full emergency contact list for their CPF is shown with no search required. List is pre-sorted by `priority`.
- Given the user's current location is known → sector-local services (nearest police, fire, hospital) pinned to top.
- Given the user taps a service → native dialer opens with `primary_number` (no confirmation dialog — the emergency *is* the confirmation).
- Given the dial-out succeeds AND the user has an active patrol → an escalation event `{patrol_id, service_id, service_name, called_at}` is appended to the patrol record as evidence of escalation.
- Given the device is offline → the last-known encrypted cache is served; banner indicates "offline — last updated X min ago".

Failure outcomes

Code	Trigger
`UNAUTHORIZED`	Not authenticated. (Even in emergencies we do not open an unauth door — the phone's native 10111 dial-out is the fallback.)
`SERVICE_NUMBER_UNVERIFIED`	`verified_at` older than 90 days → shown as a warning badge, not a block
`NO_SERVICES_CONFIGURED`	Empty state with "ask your admin to configure emergency contacts"
`SYNC_UNAVAILABLE`	Remote store offline AND no local cache → hard failure, surface native 10111 prompt

Security rules

- MUST: full contact list cached on device, encrypted at rest, refreshed on every app open when online.
- MUST: any authenticated user may view (no sector scoping — if you're helping out in a neighbour sector you still need the numbers).
- MUST: tap-to-call auto-logs to the active patrol record as an escalation event when one exists.
- MUST: service records are admin-maintained only; no patroller edits.
- SHOULD: list sorted with most-likely-needed services first (police, EMS, fire).
- SHOULD: services with `sensitive = true` hidden from non-patroller access levels.

Related: `auth/patroller-login`, `workflow/stand-down-patrol` (escalation events attach to the active patrol).

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8 — `data/live-patroller-map` (new — not in the original 8 mockups)

Problem. An active patroller — and dispatch — needs Uber-style situational awareness: *who else is out there right now, and where?* This enables backup requests, cross-patrol coordination, and faster dispatch to incidents. Distinct from `hotspots-map` (which shows *historical incidents*) — this shows *current colleagues*.

Fields

Ingestion (heartbeat):

Field	Type	Required	Notes
`patrol_id`	uuid	yes	Must reference a patrol in `state = active`

`call_sign`	string	yes	From device session
`lat`	number	yes	
`lng`	number	yes	
`heading`	number	no	Degrees
`speed`	number	no	km/h
`accuracy_m`	number	yes	GPS horizontal accuracy
`timestamp`	timestamp	yes	Device-provided; server corrects obvious drift
`signature`	string	yes	Device key signature over the heartbeat

Subscription (viewer):

Field	Type	Notes
`scope`	enum	`sector` (patrollers) or `cpf` (dispatch) — derived from viewer's access level
Response per pin	object	`{call_sign, patrol_type, vehicle, lat, lng, heading, last_update, duration_on_patrol_min}`

Success outcomes

- Given an active patroller in sector X → the map shows pins for all other active patrollers in sector X, refreshing every 30 s.
- Given a `call_centre_agent` (dispatch) → the map shows pins for all active patrollers across every sector of their CPF.
- Given a pin is tapped → call sign, patrol type, vehicle, minutes on patrol, and seconds since last heartbeat are displayed.
- Given a patroller stands down → their pin is removed from the live cache within 5 s.

Failure outcomes

Code	Trigger
`UNAUTHORIZED`	Not authenticated, or not on an active patrol AND not dispatch
`LOCATION_STALE`	Viewer sees a pin whose last heartbeat is >2 min old → pin greys out (not removed) — tells you the patroller is in a dead zone
`REALTIME_CHANNEL_DOWN`	Websocket unavailable → fall back to polling every 30 s (degraded, not an error to the user)

`HEARTBEAT_RATE_LIMITED`	A client sending heartbeats faster than 1 per 20 s is throttled — anti-spam
`NO_ACTIVE_PATROLS`	Empty map, empty-state message (not an error)

Security rules

- MUST: pin only visible while its patroller's patrol is `active`. Stood-down patrols are purged from live cache within 5 s.
- MUST: patrollers scoped to same-sector visibility. Dispatch (`call_centre_agent`) scoped to whole-CPF visibility. No cross-CPF visibility.
- MUST: heartbeat rate limited to 1 per 20 s per `patrol_id`.
- MUST: live cache is ephemeral — wiped on stand-down. Historical polyline lives on the sealed patrol record (feature 3), not in the live cache.
- MUST: heartbeats signed with the device key issued at login, so a malicious client cannot forge another patroller's location.
- SHOULD: dispatch's viewing events are audit-logged (who looked at whom, when).

Related: `auth/patroller-login`, `workflow/commence-patrol`, `workflow/stand-down-patrol`.

Follow-ups (not in this brainstorm batch)

1. **`data/member-profile-admin-edit`** — sector-scoped admin-only edit path for member records. Triggered by the Feature 6 decision "only admin and only admins of own sector".
2. **`integration/incidents-api`** — contract-only blueprint for the external data source read by `hotspots-map`. Models the request/response shape the hotspots blueprint depends on, without specifying the external service.
3. **Mockup gaps to fix in the iPhone 16 & 17 Pro designs** before implementation:
    - Screen 3 (Commence Patrol): add `Odometer start` numeric field.
    - Screen 4 (Stand Down): "KM's Traveled" becomes read-only / computed; add `Odometer end` input; add SARS-compliance indicator (✓ or ! badge); add start/end location indicators.
    - Screen 7 (Members): add on-duty badge and next-of-kin expand section.
    - **New screen**: Live Patroller Map — missing from the mockups entirely.

Self-review checklist (SKILL.md Step 8)

Check	1	2	3	4	5	6	7	8
Kebab-case name	✓	✓	✓	✓	✓	✓	✓	✓

Valid FDL category	auth	workflow	workflow	data	data	data	data	data
At least one success outcome	✓	✓ ×3	✓ ×3	✓	✓	✓	✓	✓
At least one failure outcome	6	8	8	4	5	5	4	5
Every failure has an error code	✓	✓	✓	✓	✓	✓	✓	✓
At least one security rule	✓	✓	✓	✓	✓	✓	✓	✓
`related:` populated	n/a	→ 1	→ 1, 2	→ 1	→ 1	→ 1, 8	→ 1, 3	→ 1, 2, 3
No placeholder language	✓	✓	✓	✓	✓	✓	✓	✓

All eight blueprints pass the internal completeness gate. Ready for `/fdl-create` handoff on your approval.